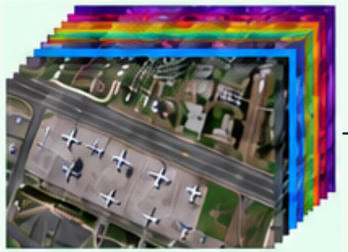
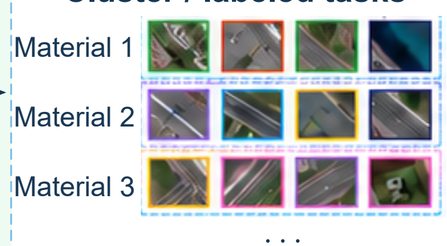
(A) Offline material-task learning

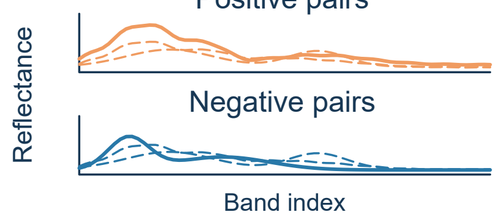


Development scenes
Multiple $H \times W \times L$ cubes

Cluster / labeled tasks



Random references + spectral augmentation



MetaMatch

Weighted BCE

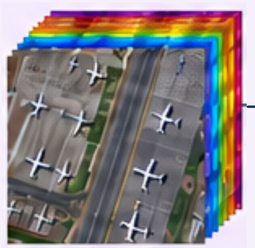
$$\mathcal{L} = -w_p y \log p - w_n (1-y) \log(1-p)$$

Train

Transfer learned weights

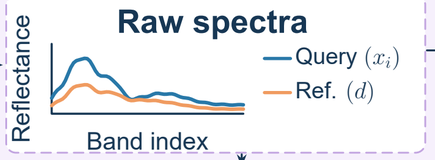
(B) Background-conditioned pair input

New image X

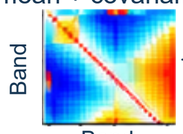


$N \times L$

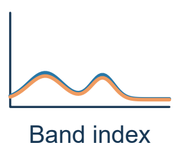
Raw spectra



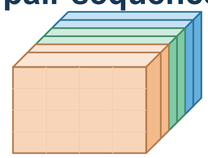
Scene statistics
mean + covariance



Whitening

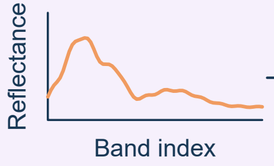


9-channel pair sequence



spectra / product
difference
band position
$9 \times L$

Reference d



4 matching statistics

s_1 s_2 s_3 s_4

(C) Shared MetaMatch network

Frozen at inference

Conv1D 9 to 64, $k = 7$

Res 1 $k = 5,\ d = 1$ — **Res 2** $k = 5,\ d = 2$ — **Res 3** $k = 3,\ d = 4$ — **Res 4** $k = 3,\ d = 8$

2 conv layers + GroupNorm + GELU

$9 \times L$ → $64 \times L$

Pooling

- Attention → 64-D
- Mean → 64-D
- Max → 64-D

192-D

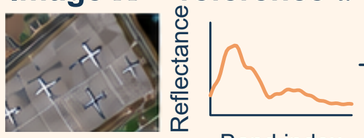
MLP 4 to 32 → 32-D

Concat → 224-D → **Head** 224 to 128 to 1 → **Sigmoid** → s_i

Variable L → fixed-size representation

(D) Fixed-weight reference adaptation (MetaMatch-TTA)

Image X + reference d



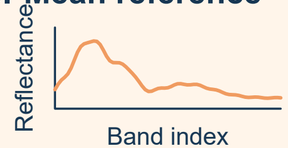
$N \times L$

1. Initial scoring — Frozen MetaMatch → $s^{(0)}$

2. Select top-k

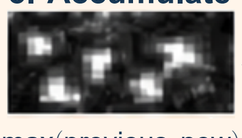


3. Mean reference



4. Rescore — Frozen MetaMatch → New scores

5. Accumulate — max(previous, new)



MetaMatch-TTA



$s^{(2)}$

Repeat steps 2-5 twice

Data flow — Shared weights (offline learning) — Training only

Scene and score thumbnails are schematic.

Optional +S

Student network

Train student from teacher scores
Average teacher and student predictions

MetaMatch-TTA +S

